Heltec e290 ChatterBox Mesh Node - DIY

The Heltec Vision e290 is an excellent base platform for a ChatterBox node. You can choose whether to build a standard node, or one with additional capabilities (radar/ proximity sensing, remote control, thermal) simply by plugging in more components.

What it Can Do

- Full ChatterBox node, with secure packet caching, display, and more.
- Track & share location
- Detect motion / distance and optionally notify your cluster
- Close relay automatically (switch) on motion detected
- Switch a circuit on/off remotely
- Report last motion

Heltec Vision e290

Required Components

- Heltec Vision e290
- 3.7 LiPo Battery + JST/PH2.0 Adapter
- Antenna & SMA Connector

Time Source and/or GNSS

- Adafruit DS3231 RTC
- SparkFun RV-8803 RTC
- SparkFun SAM-M8Q GNSS
- SparkFun SAM-M10Q GNSS

Optional

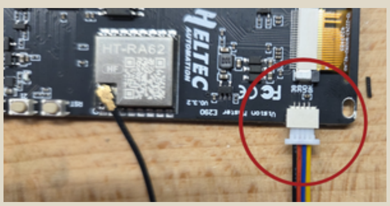
- SparkFun Qwiic Relay
- Sparkfun Pulsed Coherent Radar

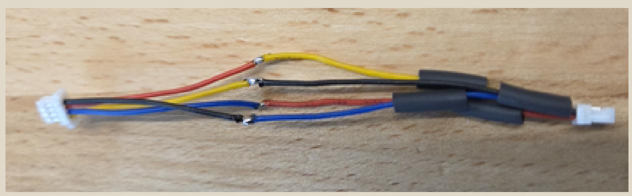
Build Summary

This build takes about 30 minutes. Once you have your chosen components, the build is essentially creating a Qwiic adapter for the e290, plugging all the components together in a chain, connecting a battery, and inserting into a case.

1. Build a Qwiic / Heltec Adapter

Heltec's plug port will physically fit a Qwiic plug, but the wiring/pins are not compatible. Make an adapter as shown.





Cut and rewire a standard Qwiic wire:

Black → Blue
Red → Yellow
Blue → Red
Yellow → Black

2. Choose a Node Base

Basic Node

Fully functional node, not dependent on GPS

e290 + DS3231

– OR –

e290 + RV-8803 RTC

Realtime Clock Options

Adafruit DS3231

SparkFun RV-8803 RTC

GPS Node

Fully functional node, regularly/securely shares its location

e290 + SAM-M8Q GNSS

– OR –

e290 + SAM-M10Q

– OR –

e290 + [either GNSS] + RV8803

GPS/GNSS Options

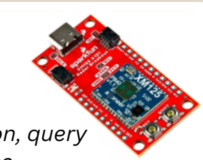
SAM-M8Q GNSS

SAM-M10Q GNSS

3. Optional - Add More

These are optional components you may or may not choose to add.

Proximity Sensor
Sparkfun Pulsed Coherent Radar



Be notified of motion, query last motion/distance

Remote On/Off Switch
SparkFun Qwiic Relay

Remotely open/close a circuit, basically an on/off switch

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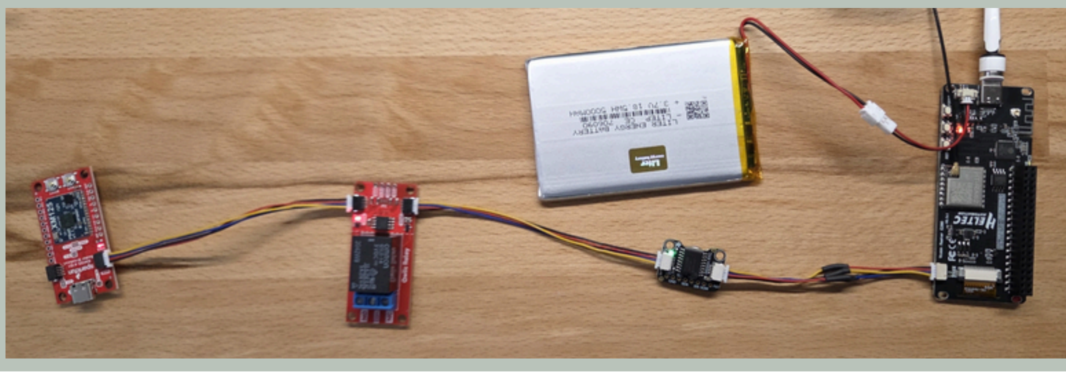
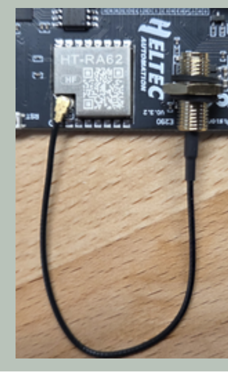
This can be dangerous, do not attempt unless you undersanding electronics!

4. Connect All Components

The selected components can be wired together in any order, in sequence. The sequence shown is one option, but if you've selected different components, yours will look different than below.

Also connect the battery (using the adapter in the parts list) and the antenna SMA connector.





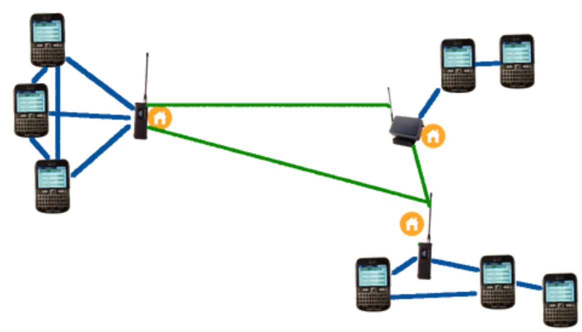
5. Install Firmware

Flash the ChatterBox node firmware (free) onto the new node by following instructions on the firmware page.

https://chatters.io/firmware

6. Onboard the Node

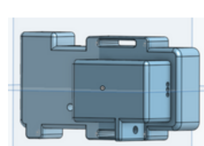
- *Power up the node*
- *On your Root communicator, select "Settings / Cluster / Onboard New Device"*
- *Wait a minute or two, it should automatically join*



Your cluster will automatically learn about (and use) the new node!

Within minutes, other devices in your cluster will automatically exchange public keys with the new node, and it should show up on neighbors screens. While existing devices are learning, it may temporarily show up as "unidentified".

This can be dangerous, do not attempt unless you undersanding electronics!



3D Printable Case Options Coming Soon

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